# Real-robot run, 28 October 2026

Machine Minds · FANUC ER-4iA cell · 16:34–17:16 · request q69cdbc, run r20261028-716eb3 · policy MachineMinds-v5-ClosedLoop, weights `yolo_seg_v5_real_1080` · written from the robot's own logs, camera images and the results dashboard

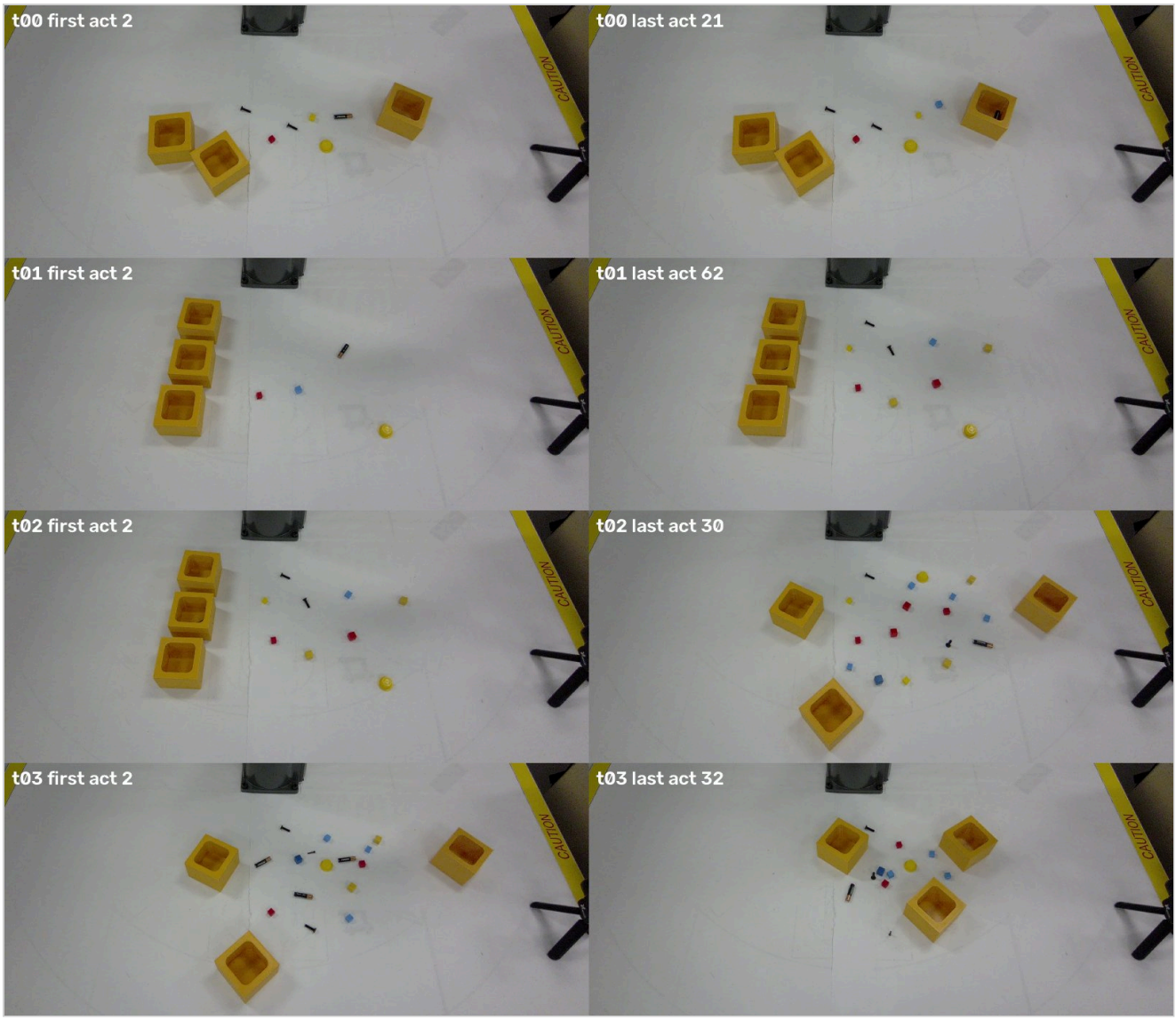
**The short version**

- **Official score: 4 of 8 rollouts marked success (50 %, 95 % interval 22–78 %), 4 faults.** 8 of the 10 requested rollouts were run; mean cycle 293 s. The dashboard still shows the run as "in progress".
- **Detection is fixed.** Yesterday v4 found almost none of the real objects. Today the fine-tuned model found every bin, cube, screw, AAA cell and cap in every picture, with confidence around 0.8–0.9.
- **The robot only sorted AAA cells (and one cube).** Every other cube and every screw was refused by the **pose-fit check**, the step that works out an object's exact position and angle before a grasp. With nothing it was allowed to pick, the robot waited ("hold") until the cell ended the trial.
- **The scores need a second look.** In the four trials marked success, our own camera images show cubes and screws still on the table at the end, which does not meet the task's success rule ("every target is in a bin"). The organizers may revise these.

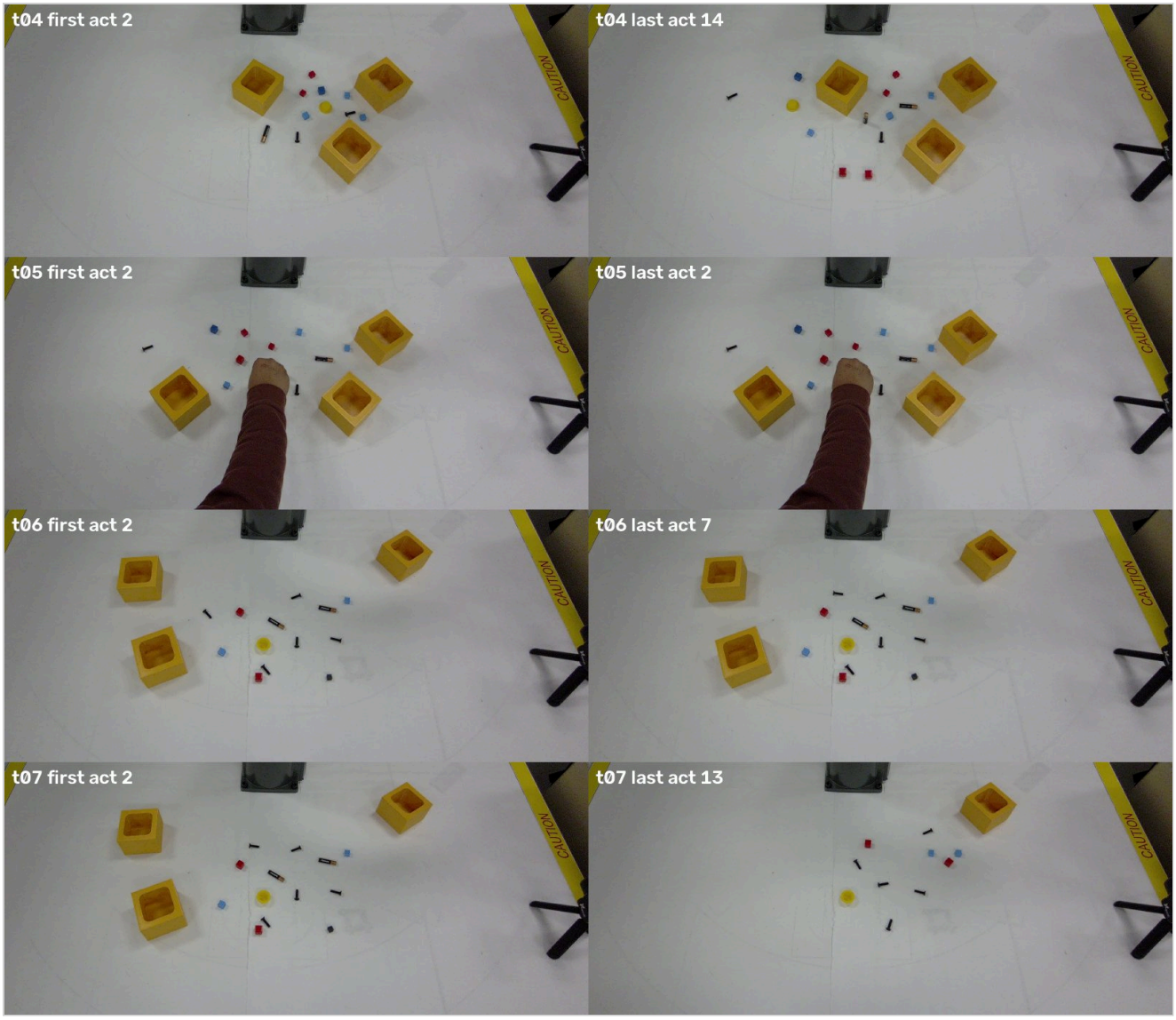
## 1. Rollout by rollout

| Rollout | Official | Length | What the robot did | How it ended |
|---|---|---|---|---|
| t00 | success | 3 min 52 s | Put 1 AAA cell in its bin (confirmed in the next picture). 2 cubes and 2 screws refused by the pose-fit check; held for 17 acts. | Cell ended the trial (robot homing). |
| t01 | success | 5 min 28 s | Put 1 AAA cell in its bin. Held for 56 acts. The table had 1 AAA cell, 2 cubes and the cap at the start; more cubes and screws appeared on it during the trial. | Cell ended the trial. |
| t02 | success | 6 min 4 s | Held 4 acts at the start, then put 2 AAA cells in the bin. Held 14 acts. The bins were moved during the trial and a third AAA cell appeared; the robot went for it. | Cell ended the trial during that pick. |
| t03 | fault | 8 min 30 s | Put 3 AAA cells in the bin. A person then rearranged the table by hand (taking the cells back out); the robot put 2 of them back. | Ended during the 6th AAA pick; bins had been moved. |
| t04 | fault | 3 min 15 s | Put 1 AAA cell in the bin, held 7 acts, then went for a second AAA cell. | Ended during that pick. |
| t05 | fault | 44 s | A person's arm was on the table in the first picture. The robot started an AAA pick. | Ended during that pick. |
| t06 | fault | 2 min 34 s | First cube pick of the day (blue cube, pose fit 0.92). The next picture showed it still there ("missed"); the robot tried again. | Ended during the retry. |
| t07 | success | 4 min 31 s | Picked the same blue cube and put it in the cube bin (confirmed). Then 3 picks were refused because the path left the cell's allowed TCP box, and it held. | Cell ended the trial; by then 2 bins had been taken off the table. |

Picks confirmed in the logs (gone from their spot in the next picture): 10 AAA cells and 1 cube; 1 cube missed. Every pick was aimed at the bin for its type, and the cap was never picked. The cause of the 4 faults is not in our logs. All four ended while a pick motion was under way; the dashboard videos should show what the cell saw.




**t00–t03, first and last picture of each trial.** In t00 the cubes and screws are still on the table at the end. In t01–t03 the scene changes during the trial: objects are added and the bins are moved.



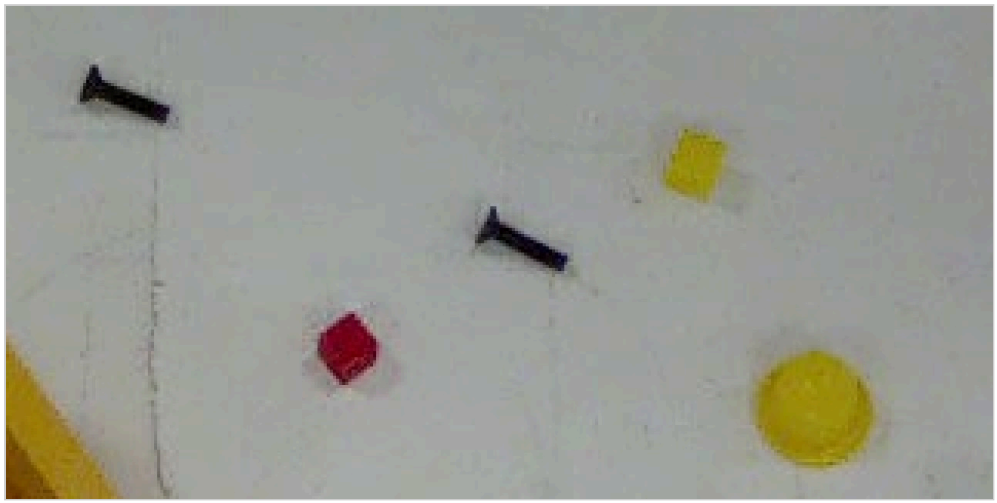

**t04–t07.** In t05 a person's arm is on the table when the robot takes its first picture. In t07 the blue cube was sorted; at the end two bins have been removed.

## 2. Why the robot left the cubes and screws

For each object the program goes through four steps:

1. **YOLO** finds the object, names it and outlines it (the "mask"). *This worked today.*
2. **Pose fit** places a 3D model of the object (e.g. a 10 mm cube) at many positions and angles, draws what the camera would see, and keeps the best match. The match score is the overlap between that drawing and YOLO's outline: 1.0 is identical.
3. **The gate** refuses any object whose best match is under **0.85**, rather than risk a bad grasp.
4. **The planner** only picks objects that passed. If none did, it holds.

| Object | Detections (all pictures) | Lowest | Median | Highest | Passed the 0.85 gate |
|---|---|---|---|---|---|
| AAA cell | 36 | 0.85 | 0.90 | 0.93 | 36 (100 %) |
| cap (distractor) | 122 | 0.76 | 0.95 | 0.96 | not picked by design |
| screw | 163 | 0.42 | 0.76 | 0.79 | 0 (0 %) |
| cube | 694 | 0.39 | 0.55 | 0.92 | 3 (0.4 %), all the same blue cube |



**The real camera picture, enlarged 4× with no smoothing.** Each 10 mm cube is only about 20 pixels across, and its edges are blurred over several pixels. An outline just 1 pixel too large on every side already scores 400 ÷ 484 = 0.83, under the gate, even if the position and angle are exactly right. AAA cells and the cap are much larger in the picture, so the same error costs them little.

The gate was set in simulation, where cubes and screws score above 0.85. The checks before the slot (one simulated layout at real arm speed, and the mock cell) never measured the pose fit on real pictures, so they did not catch this.

What is not known yet is **which part** causes the low scores: YOLO's outlines being too loose at this size, the camera calibration drawing the model slightly off, the 3D model's dimensions, or simply a gate too strict for 20-pixel objects. The cube scores fall into two groups (about 0.40–0.55 and 0.62–0.80), which suggests more than one cause. Today's saved pictures are enough to answer this offline.

## 3. Other things seen today

- **The table changed during trials.** Objects were added, bins were moved and, in t07, removed. The program names the bins from the first picture and then follows them, so moved bins can make later paths fall outside the allowed TCP box (as in t07).
- **People in the workspace.** A hand or arm was on the table in t03 and t05 while the robot was working. Our program does not check for people; it relies on the cell's own safety system.
- **Holding until timeout wastes time.** When targets remain but none passes the gate, the robot holds until the cell gives up, 3–6 minutes per trial. That is why only 8 of the 10 rollouts fit into the slot.

> **What went right:** the fine-tuned model detected everything on the real table; 10 AAA cells and a cube were picked, each aimed at the bin for its type; and it never picked the cap.

## 4. Next steps (nothing changed yet)

1. **Diagnose the pose fit on today's pictures:** overlay YOLO's outline and the fitted model on cube and screw crops, and find which part is off (outline, calibration, model size or gate). Fix that part and check it on these pictures and in simulation before the next slot. Lowering the gate just to pass is not the plan.
2. **Decide what the robot does when nothing passes the gate**, instead of holding until timeout.
3. **Ask the organizers** about the 4 faults (the videos show the cause) and about the success marks for t00–t02 and t07.

Sources: `v5\runs\real\20261028_163407\rollout_02-09` (t00–t07): act_*.json and table pictures; client output in the terminal; results dashboard read at 18:15. Not verified: fault causes, and whether the dashboard scores are final.